



SOCIAL MEDIA IMPACT ON MENTAL HEALTH SCORE

Exploratory Data Analysis and Statistical
Investigation

Research question:

Is higher daily social media use associated with lower mental health scores?

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Social Media Impact on Mental Health Score

Introduction

The following report on a Student Health & Grades dataset focuses on exploratory analysis and statistical investigation. The exploration will pertain to a potential relationship between daily social media usage and lower mental health scores, with the underlying research question: *is higher daily social-media use associated with lower mental health scores?* Therefore, the alternative hypothesis (H_1) may be formulated as: *“There is a negative linear association between daily social media usage and mental health scores”*.

Dataset

The dataset used thereafter, was sourced from [Kaggle Datasets](#), under the CC0: Public Domain licence and created by user *Debayan Bandyopadhyay*. The dataset is purely synthetic and does not contain any real-world human data. Therefore, the data may not be representative of real students, and relationships found during the analysis should not be interpreted as evidence for the same relationships existing among real students.

The dataset consists of 15,000 records and 14 variables (see Appendix A, Figure A.1). For the following investigation, *Mental_Health_Score* and *Daily_Social_Media_Hours* are of particular interest and will be thoroughly analysed. Initial exploration revealed no missing values or duplicate records.

Exploratory Data Analysis

The initial analysis was conducted to examine potential relationships between mental health and social-media usage. The exploratory data analysis (EDA) focused on the distribution of the *Mental_Health_Score* and *Daily_Social_Media_Hours* variables, identifying patterns that may be relevant to the established hypothesis

Histogram Plot

The *Mental_Health_Score* and *Daily_Social_Media_Hours* data was visualised using a histogram. The mental health score distribution centred around the 60-90 range (see Appendix A, Figure A.2) while daily social media usage histogram revealed the distribution concentrated approximately at 2-8 (see Appendix A, Figure A.3).

Scatter Plot

A scatter diagram was used to visually examine potential relationships, variation and movement trends before calculating any statistical measure. As a consequence, the most observations were concentrated between 60-90 range in the mental health score at the 2-8 hours range of daily social media usage (see Appendix A, Figure A.4). The centre of the cloud appears to move downward, while becoming more spread out. Thus, the scatter plot suggests a negative

association between daily social-media usage and mental health score while appearing to show some variation.

Pearson's Correlation

Utilising the Pearson's correlation coefficient (*Schober et al.*) produced $r = -0.3998$ and $p < 0.001$ (see Appendix A, Figure A.5). Such findings suggest a moderate negative linear association between social media usage and mental health score. This indicates *Daily_Social_Media_Hours* tends to be associated with lower *Mental_Health_Score*. However, correlation does not demonstrate causality of lower mental health scores.

Statistical Significance

To investigate the statistical significance of the moderate negative linear association between *Mental_Health_Score* and *Daily_Social_Media_Hours*, a null hypothesis test (H_0) may be formulated as: “*There is no linear relationship between social-media usage and mental-health score.*”.

The very small p-value indicates that a correlation of this magnitude would be highly unlikely under the null hypothesis of no linear association. However, causation cannot be established based on statistical significance of the $r = -0.40$ correlation. The simple linear regression has an R^2 of approximately 0.16, meaning 16% of the variance in *Mental_Health_Score* is explained by the linear model using *Daily_Social_Media_Hours* as the predictor (see Appendix A, Figure A.6).

Investigating other variables within the dataset reveals significant linear association with mental health scores. In particular, $r = +0.311$ for *Sleep_Hours*, suggests that as sleep increases, *Mental_Health_Score* tends to increase. Similarly, $r = +0.246$ for *Physical_Activity_Hours* is indicative of a weak positive linear association albeit, the relationship is relatively noisy. Moreover, *Social_Isolation_Score* seems to be associated with lower *Mental_Health_Score*, since the correlation of $r = -0.292$ suggests a weak negative relationship (see Appendix A, Figure A.7).

Practical Significance

To examine the practical significance of this correlation, the students were divided into above and below average social media users. Thus, the mean for low social media users was 75.46 with standard deviation at 7.96 while mean for high social media users was 69.50 with standard deviation at 9.50 (see Appendix A, Figure A.8). Despite the groups having different averages, there is substantial overlap between them. However, it is unclear whether 6-point difference in averages is large compared with natural variation existing in mental health scores to conclude the result practically significant.

Cohen's d statistical measure reveals a substantial gap in mental health scores between the two group averages. Specifically the high social media group scored an average of 0.679 standard deviations lower than the low social media group, indicating a moderate-to-large standardised difference.

Linear Regression Analysis

Simple Linear Regression

Simple linear regression estimated the relationship between *Daily_Social_Media_Hours* and the *Mental_Health_Score*. As a result, the predicted *Mental_Health_Score* decreases by approximately 1.54 points for each additional hour of social media usage. The standard error of the slope is 0.029, describing how much uncertainty there is around the estimated slope. The *r* value is -0.40 suggesting a moderate negative linear relationship, which was previously independently verified using the Pearson's correlation coefficient.

Multiple Linear Regression

Considering *Sleep_Hours*, *Physical_Activity_Hours* and *Social_Isolation_Score* simultaneously using multiple linear regression (Northwestern University) will investigate whether *Daily_Social_Media_Hours* still predicts *Mental_Health_Score* under such conditions. In the multiple regression model, the coefficient for *Daily_Social_Media_Hours* was -1.154, becoming less negative once the other variables were included (see Appendix A, Figure A.9). 1.15 point lower mental health score may be associated with each additional hour of social media use, while holding *Sleep_Hours*, *Physical_Activity_Hours* and *Social_Isolation_Score* constant. *Daily_Social_Media_Hours* remained significantly negatively associated with *Mental_Health_Score* after controlling for sleep, physical activity and social isolation.

Holding the other variables constant, each additional hour of sleep was associated with approximately 1.5 point higher *Mental_Health_Score*. Similarly, each additional hour of physical activity was associated with approximately a 1.59-point increase in *Mental_Health_Score*. While the effect is smaller than the coefficient for other variables, *Social_Isolation_Score* sees an approximately 0.21-point decrease in *Mental_Health_Score*. Moreover, the R^2 increased to 0.235, meaning the multiple linear regression model explains approximately 23.5% of the variation in *Mental_Health_Score*.

The coefficient decreased in magnitude by approximately 25% after including *Sleep_Hours*, *Physical_Activity_Hours* and *Social_Isolation_Score*, suggesting that some of the association between *Daily_Social_Media_Hours* and *Mental_Health_Score* overlaps with variation captured by other predictors

Residuals

Upon examining the Scatter plot, the initial indication is encouraging as the residuals appear to centre around zero (see Appendix A, Figure A.10). The points are distributed both above and below the horizontal zero line, with no obvious curve or systematic pattern. However, across the range of predicted *Mental_Health_Score*, there appears to be a larger vertical spread. The spread narrows towards the higher scores, creating an elongated funnel-like shape, suggesting the residual variance is not completely constant, potentially relating to heteroscedasticity.

Bruesch-Pagan Test

The Breusch-Pagan test (*ten Tije*) provided strong evidence for heteroscedasticity ($p < 0.001$) (see Appendix A, Figure A.11). This appears to match the spread across the range of predicted *Mental_Health_Score* in the residuals Scatter plot. Due to 15,000 observations in the dataset, the Breusch-Pagan test possesses considerable statistical power to detect relatively small departures from constant variance. Therefore, conventional OLS standard errors may be unreliable.

Residuals Histogram

The observed residual distribution is not perfectly normal, with most residuals concentrated approximately between the -10 and +10 range. The distribution is not symmetrical around zero, with a negative tail extending towards -35 to -40 and positive tail only extending to around +20 to +25 (see Appendix A, Figure A.12).

Q-Q Plot

The Q-Q plot follows the reference line reasonably close through the central quantiles, but shows departure in both tails, indicating that the residuals are approximately normal in the centre but deviate from normality at the extremes (see Appendix A, Figure A.13).

HC3

HC3 heteroscedasticity-robust standard errors (*Rajh-Weber et al.*) were used to provide more reliable estimates of coefficient uncertainty in the presence of heteroscedasticity. This allows the resulting t-statistics, p-values or confidence intervals to be assessed using standard errors that are robust to heteroscedasticity. HC3 in particular applies strong adjustment for observations with high leverage and is commonly recommended as a conservative choice.

Upon applying HC3 it may be observed that the detected heteroscedasticity does not materially change the conclusion of the regression. As expected, the coefficients have not changed, only the estimates of the uncertainty surrounding them (see Appendix A, Figure A.14). Sleep and physical activity remain positively associated with mental health, while social media usage and social isolation maintained their negative association. The predictors remain statistically significant, and the substantive conclusions of the model survived the robust standard error correction.

Conclusion

The analysis found a moderate negative linear association between *Daily_Social_Media_Hours* and *Mental_Health_Score* ($r = -0.400$, $p < .001$). This association remained statistically significant in the multiple regression model after accounting for *Sleep_Hours*, *Physical_Activity_Hours* and *Social_Isolation_Score*, where each additional hour of social media usage was associated with approximately a 1.15-point lower *Mental_Health_Score*. The full model explains 23.5% of the variation in *Mental_Health_Score*.

Cohen's $d \approx -0.679$ indicates a moderate-to-large standardised difference between high and low social media use groups with the high-use group having the lower *Mental_Health_Score*.

HC3 heteroscedasticity-robust standard error was utilised when heteroscedasticity was identified during diagnostics. The residuals also showed some departure from normality. The resulting inference was largely unchanged, suggested that primary conclusions were robust to the detected heteroscedasticity.

Overall, the evidence supports the hypothesis within this dataset. Higher *Daily_Social_Media_Hours* is associated lower *Mental_Health_Score*. This conclusion is supported by the negative Pearson's correlation, simple and multiple regression coefficients, the observed group difference and the fact that the regression inference remained essentially unchanged after applying HC3 robust standard errors.

However, because the analysis is observational and due to the synthetic nature of the dataset, the results demonstrate an association within the dataset, rather than evidence for causation or evidence that the same relationship exists among real students.

References

Bandyopadhyay, Debayan. “AI & Social Media Impact: Student Health & Grades.” *Kaggledatasets*, 3 Aug. 2026, <https://www.kaggle.com/datasets/debayank2024/ai-and-social-media-impact-student-health-and-grades>. Accessed 16 Aug. 2026.

Matplotlib Developers. “API Reference — Matplotlib 3.5.0 Documentation.” *Matplotlib.Org*, <https://matplotlib.org/stable/api/index>. Accessed 16 Aug. 2026.

Northwestern University. “Linear Regression Resource Guide — Northwestern IT Research Computing and Data Services Northwestern University.” *Northwestern.Edu*, 24 July 2025, <https://rcdsdocs.it.northwestern.edu/guides/stats/linear-regression.html>. Accessed 16 Aug. 2026.

NumPy Developers. “NumPy Reference — NumPy V1.23 Manual.” *Numpy.Org*, 28 June 2026, <https://numpy.org/doc/stable/reference/index.html#reference>. Accessed 16 Aug. 2026.

pandas Developers. “API Reference — Pandas 1.1.4 Documentation.” *Pandas.Pydata.Org*, <https://pandas.pydata.org/docs/reference/index.html>. Accessed 16 Aug. 2026.

Perktold, Skipper Seabold. “Statsmodels: Econometric and Statistical Modeling With Python.” *Proceedings of the 9Th Python in Science Conference*, 2010.

Rajh-Weber, Hanna, et al. “A Practice-Oriented Guide to Statistical Inference in Linear Modeling for Non-Normal or Heteroskedastic Error Distributions.” *Behavior Research Methods*, vol. 57, no. 12, 10 Nov. 2025, pp. 338–338, 10.3758/s13428-025-02801-4.

Schober, Patrick, et al. “Correlation Coefficients: Appropriate Use and Interpretation.” *Anesthesia & Analgesia*, vol. 126, no. 5, May 2018, pp. 1763–1768, https://journals.lww.com/anesthesia-analgesia/fulltext/2018/05000/correlation_coefficients__appropriate_use_and.50.aspx, 10.1213/ANE.0000000000002864. Accessed 16 Aug. 2026.

SciPy Developers. “SciPy Documentation — SciPy V1.8.1 Manual.” 19 June 2026.

ten Tije, Matthijs . “Tilburg Science Hub.” *Tilburgsciencehub.Com*, 2020, <https://www.tilburgsciencehub.com/topics/analyze/tests/tests-assumptions/heteroskedasticity/>. Accessed 16 Aug. 2026.

APPENDIX A – SUPPORTING ANALYSIS

The following figures and statistical outputs provide supporting evidence for analysis presented in the report. The figures are presented in the order in which they appear throughout the report.

A.1 Dataset Overview

Figure A.1 – Dataset Overview

A.2 Exploratory Data Analysis

Student_ID	Age	Gender	Education_Level	Daily_Social_Media_Hours	Daily_AI_Tool_Use_Hours	Sleep_Hours	Physical_Activity_Hours	Mental_Health_Score	Physical_Health_Score	Social_Isolation_Score	Burnout_Level	Academic_Performance_Score	Academic_Failure_Risk
STU_00001	19	Male	College	5.32	1.21	7.3	1.41	74.06	98.9	4.05	Low	88.8	0
STU_00002	16	Non-binary	High School	5.21	3.89	7.81	2.48	76.13	89.94	4.65	Moderate	50.6	0
STU_00003	25	Male	University	3.61	2.65	6.34	0.06	65.01	76.75	5.28	Low	96.67	0
STU_00004	23	Male	University	2.93	1.43	6.12	1.37	74.67	87.38	3.66	Low	85.65	0
STU_00005	20	Non-binary	College	6.52	1.64	5.23	1.14	67.74	88.04	3.25	Moderate	71.84	0
STU_00006	25	Female	University	6.04	3.67	6.4	0	59.82	78.03	6.31	High	82.74	0
STU_00007	17	Male	High School	1.98	1.66	7.03	4.44	83.05	97.97	2.55	Low	69.7	0
STU_00008	19	Female	College	3.42	5.1	4.82	0	78.76	85.95	3.7	Moderate	69.98	0
STU_00009	22	Female	University	7.85	0.7	7.12	0.4	72.6	82.53	7.05	Low	97.35	0
STU_00010	15	Female	High School	6.96	1.6	5.78	1.67	71.83	91.23	5.51	Low	84.39	0
STU_00011	19	Male	College	1.57	0.83	9.02	2.04	72.67	92.78	2.09	Low	94.77	0
STU_00012	23	Female	University	4.53	0.31	6.6	3.44	82.7	97.33	3.74	Low	61.74	0
STU_00013	23	Female	University	5.55	1.07	5.27	0.38	77.62	90.24	4.3	Low	79.25	0
STU_00014	20	Male	College	6.98	1.58	6.62	1.88	70.01	73.45	4.04	Moderate	76.26	0
STU_00015	17	Female	High School	4.82	1.74	7.86	0.64	69.99	91.66	3.86	Low	77.66	0

Figure A.2 - Mental Health Score Distribution

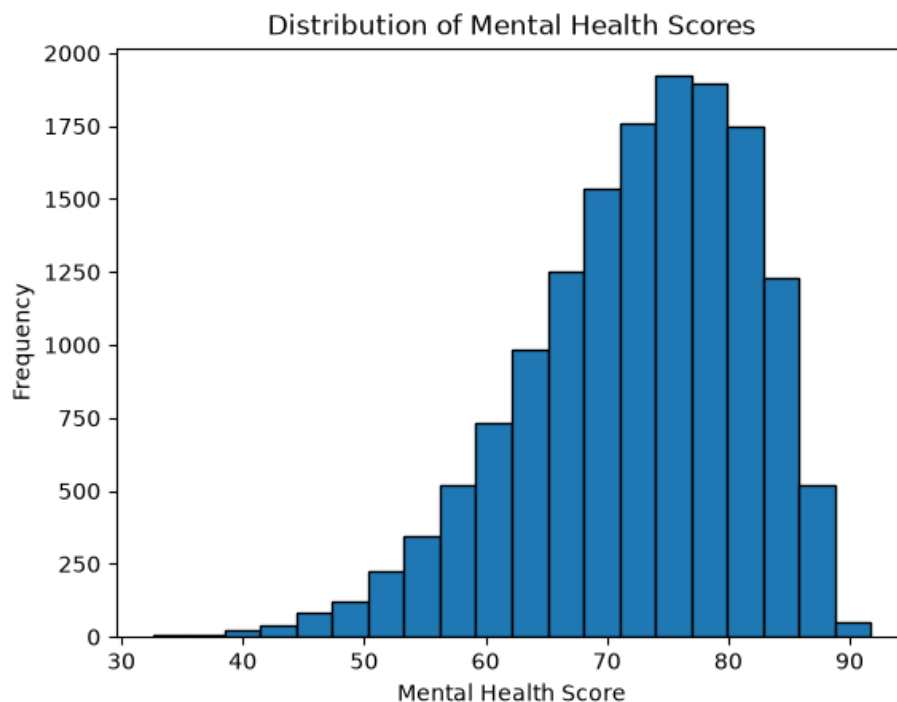


Figure A.3 – Daily Social Media Hours Distribution

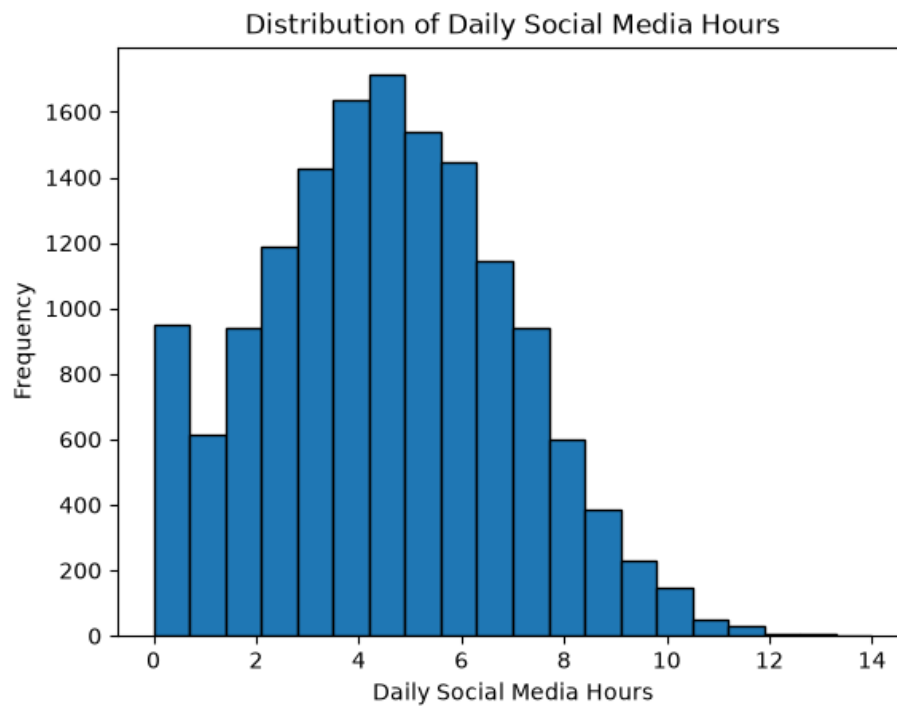


Figure A.4 — Social Media Hours vs Mental Health Score

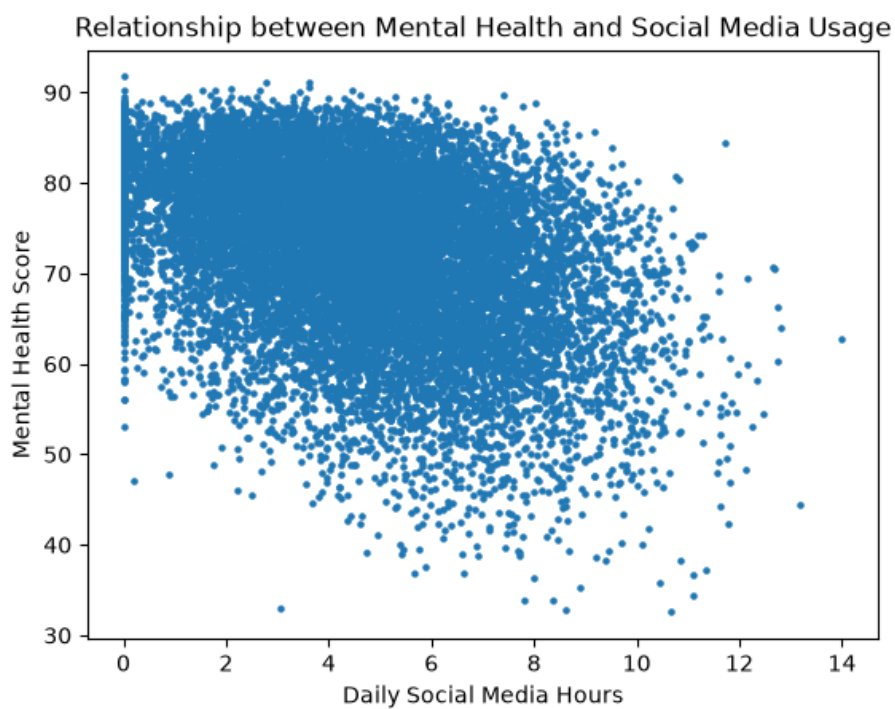


Figure A.4 shows a general negative relationship between daily social media usage and mental health score. Although considerable variation is present, students with higher levels of social media use generally tend to have lower mental health scores.

A.3 Correlation Analysis

Figure A.5 — Pearson Correlation

Correlation: -0.40
P-value: 0.000000000000000000e+00

Figure A.6 — Simple Linear Regression

Linear Regression Result:
slope=-1.541
intercept=79.506
rvalue=-0.400
pvalue=0.000e+00
stderr=0.029
intercept_stderr=0.148

Figure A.7 — Correlations with Other Variables

Correlation between Daily Social Media Hours and Mental Health Score: -0.400
Correlation between Sleep Hours and Mental Health Score: 0.311
Correlation between Social Isolation Score and Mental Health Score: -0.292
Correlation between Physical Activity Hours and Mental Health Score: 0.246

A.4 Practical Significance

Figure A.8 — High vs Low Social Media Usage

High social media group size: 7414
Low social media group size: 7586
Mean mental health score for high social media users: 69.503
Mean mental health score for low social media users: 75.455
High social media SD: 9.496
Low social media SD: 7.962
Cohen's d: -0.679

A.5 Multiple Linear Regression

Figure A.9 — Multiple Linear Regression Results

Multiple Linear Regression Results: Mental Health Score

Category	Variable / Metric	Coefficient	Std. Error	t-statistic	P-value	95% CI Lower	95% CI Upper
Model Fit	R-squared	0.235					
	Adjusted R-squared	0.235					
	F-statistic	1154.000					
	Prob (F-statistic)	0.000					
	Observations	15,000					
Intercept	const	66.8474	0.562	118.989	0.000	65.746	67.949
Predictors	Daily_Social_Media_Hours	-1.1535	0.031	-36.926	0.000	-1.215	-1.092
	Sleep_Hours	1.5008	0.056	26.988	0.000	1.392	1.610
	Social_Isolation_Score	-0.2110	0.070	-3.012	0.003	-0.348	-0.074
	Physical_Activity_Hours	1.5948	0.070	22.759	0.000	1.457	1.732
Diagnostics	Omnibus	711.227					
	Durbin-Watson	2.010					
	Jarque-Bera (JB)	818.179					
	Skew	-0.550					
	Kurtosis	3.315					

Figure A.9 Shows all variables have a p-value < 0.05, meaning they are statistically significant predictors of mental health scores. More sleep and more physical activity are associated with higher mental health scores. More social media hours and higher social isolation scores are associated with lower mental health scores. The R^2 of 0.235 indicates that about 23.5% of the variance in mental health scores is explained by these four factors.

A.6 Regression Diagnostics

Figure A.10 — Residuals vs Predicted Values

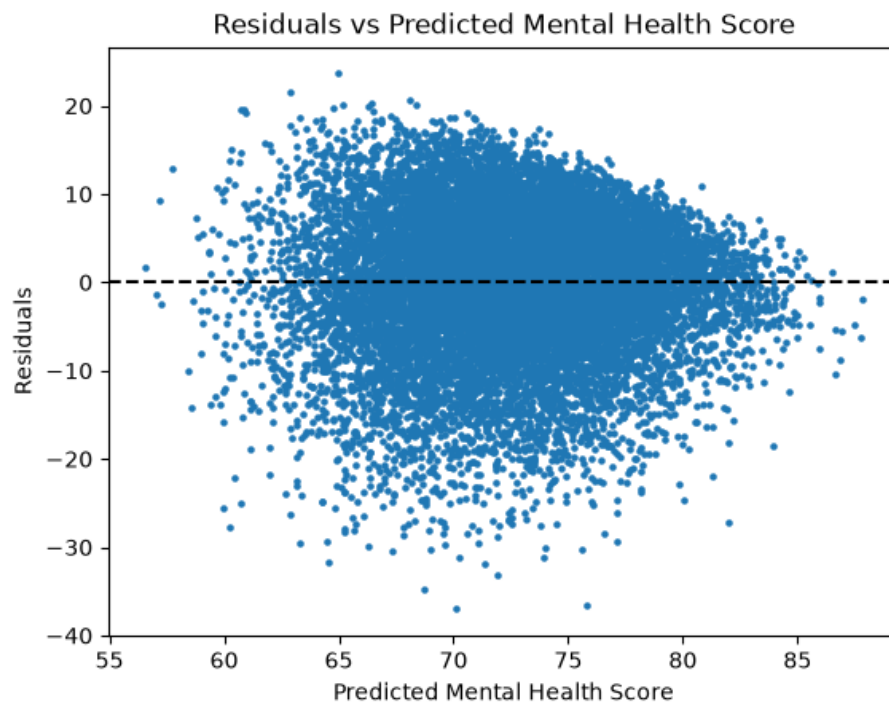


Figure A.11 — Breusch-Pagan Test

Breusch-Pagan Test Result:
LM Statistic: 379.485
LM p-value: 0.000
Robust Model Summary:

Figure A.12 — Residual Distribution

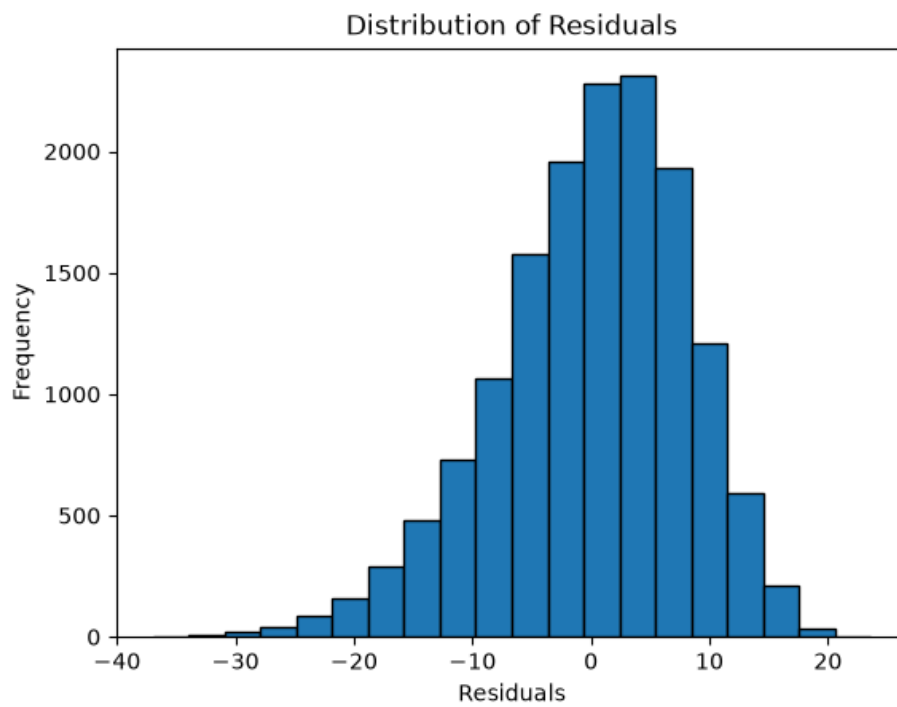
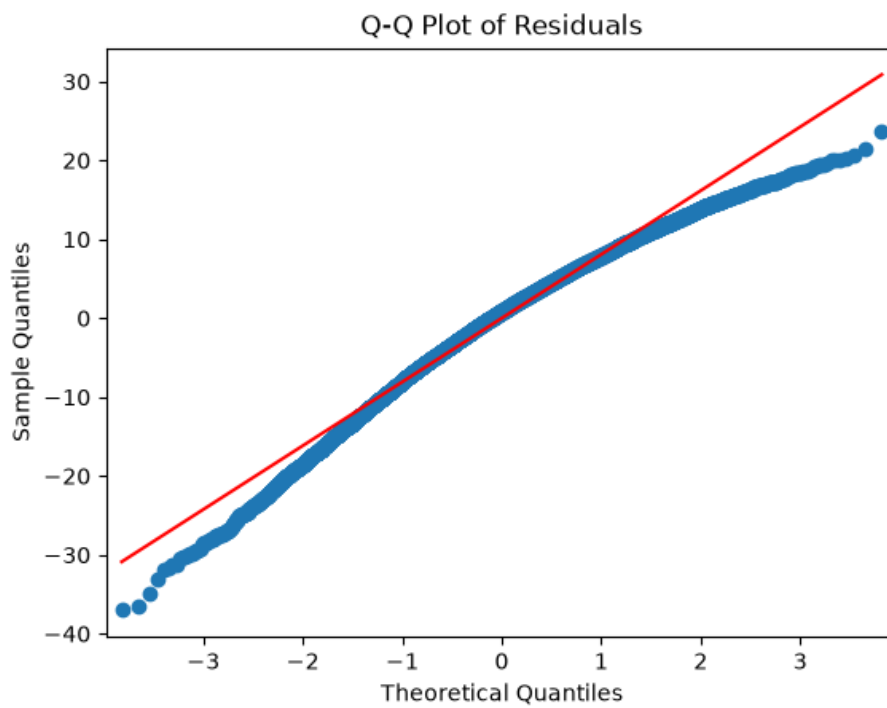


Figure A.13 — Q-Q Plot



A.7 Robust Standard Errors

Figure A.14 — OLS vs HC3 Comparison

Predictor	OLS β	OLS SE	OLS p	HC3 SE	HC3 p
Daily Social Media Hours	-1.1535	0.031	<0.001	0.031	<0.001
Sleep Hours	1.5008	0.056	<0.001	0.055	<0.001
Social Isolation Score	-0.2110	0.070	0.003	0.071	0.003
Physical Activity Hours	1.5948	0.070	<0.001	0.068	<0.001

Figure A.14 shows that HC3 robust standard errors were used to account for heteroscedasticity. The coefficients are identical, while the HC3 standard errors change only slightly, leaving the statistical conclusions unchanged.